[**E1.10 (Maths 1) 2012**]

B.ENG. and M.ENG. EXAMINATIONS 2012

PART I : MATHEMATICS 1 (ELECTRICAL AND INFORMATION SYSTEMS ENGINEERING)

Date Thursday 7th June 2012 10.00 - 12.00

DO NOT OPEN THIS PAPER UNTIL THE INVIGILATOR TELLS YOU TO.

Answer Question 1 and THREE of the remaining five

Answer Section A and Section B in different answerbooks.

Question 1 carries twice the marks of each of the other questions.

CALCULATORS MAY **NOT** BE USED.

A mathematical formulae sheet is provided.

[*Before starting, please make sure that the paper is complete; there should be 6 pages, with a total of SIX questions. Ask the invigilator for a replacement if your copy is faulty.*]

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SECTION A

[E1.10 (Maths 1) 2012]

1. (i) Write

$$z^2 + z^{*2} = -8$$

in Cartesian form.

Given that $z = x + iy$, *solve the above equation for all possible values of* y *in terms of* x.

(ii) Find all possible values of

$$(2 - 2i)^{2/3}$$

in polar form.

(iii) Find q so that the limit

$$\lim_{x\to\infty} e^{qx}\left(e^{2x/3} - (2 + e^x)^{2/3}\right)$$

is finite and non-zero.

Do not use L'Hôpital's rule.

(iv) Find the limit

$$\lim_{x\to 0} \frac{\cos(x) - 1}{x^2}$$

Do not use L'Hôpital's rule.

You can use $\lim_{x\to 0} \sin(x)/x = 1$.

(v) Differentiate

$$(\sin(x))^{\sin(x)}$$

(vi) Integrate

$$\int \frac{\sin(x) + 1}{\sin(x) - 1}\, \mathrm{d}x$$

(vii) Integrate

$$\int_0^1 \frac{x}{(1 - x^2)^{1/3}}\, \mathrm{d}x$$

Q1 CONTINUES ON THE NEXT PAGE

(viii) Find the Taylor expansion of

$$\frac{\ln(x-1)}{x}$$

about $x = 2$ to first order (up to and including the term linear in x) and state the remainder term $R_2(x)$.

(ix) Find the general solution of the following first order ODE:

$$y'(x) = \frac{y(x)}{x} + 1$$

(x) Find the general solutions of the following second order ODE:

$$y''(x) + 4y'(x) + 4y(x) = \exp(x)$$

SECTION B

2. Find $\dfrac{dy}{dx}$ as a function of x in each of the following cases :

(i) $y = \dfrac{\exp(x)}{\exp(x^2)}$;

(ii) $y = \exp(\sin^{-1}(x))$;

Note: $\sin^{-1}(x)$ *denotes the inverse* sin *function*

(iii) $y = \ln\left(\cos\left(x^{\sin(x)}\right)\right)$;

(iv) $\cos(y) = \sin(x)$;

Find the following nth derivative

(v) $\dfrac{\mathrm{d}^n}{\mathrm{d}x^n}\left(x^2 \exp(x/2)\right)$.

3. Evaluate the following limits *without using L'Hôpital's rule unless specified*:

(i) $\lim\limits_{x\to -1} \dfrac{(x-2)(x+2)}{(x-3)(x-1)}$;

Do not use L'Hôpital's rule.

(ii) $\lim\limits_{x\to -1} \dfrac{(x-2)(2x^2-2)}{(x-3)(x+1)}$;

Do not use L'Hôpital's rule.

(iii) $\lim\limits_{x\to \infty} \sqrt{x}\left(\sqrt{1+x}-\sqrt{x}\right)$;

Do not use L'Hôpital's rule.

(iv) $\lim\limits_{x\to 1} \dfrac{x^n-1}{x-1}$ for integer $n > 0$;

Do not use L'Hôpital's rule.

(v) $\lim\limits_{x\to \pi} \dfrac{\sin(x)+(x-\pi)}{(x-\pi)^3}$;

Use L'Hôpital's rule here.

PLEASE TURN OVER

4. (i) State whether the improper integral $\int_0^1 x\ln(x)\ \mathrm{d}x$ is finite and calculate its value if it is.

   Note: $\lim_{x\to 0} x\ln(x) = 0$.

   (ii) Show that $\int_0^\infty x^n \exp(-x)\ \mathrm{d}x \ = \ n\int_0^\infty x^{n-1}\exp(-x)\ \mathrm{d}x$ for any positive integer n, i.e. $0 < n \in \mathbb{N}$

   (iii) Integrate $\int \cos^3(x)dx$.

   (iv) Integrate $\int \dfrac{x^2-1}{x^3+2x^2-3x}dx$.

5. (i) Write the following equation in Cartesian form and then solve it for all possible values of y in terms of x :

$$|z+1| = z^* + z$$

   (ii) Express in polar form

$$1+2i \qquad \text{and} \qquad (\sqrt{3}+i)^{1/3}$$

   (iii) Express $\sin(4\theta)$ in terms of $\sin(\theta)$ and $\cos(\theta)$.

   (iv) Integrate $\int \cos(x)\exp(2x)\mathrm{d}x$.

   Note: You can use $\int \exp(ax)\,\mathrm{d}x = \dfrac{1}{a}\exp(ax) + C$ *for complex* $a \in \mathbb{C}$.

6. (i) Find the solution $y(x)$ of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y^2 + x^2 + 2(x+y) + 2}{2x^2 + 4x + 2}$$

Note: It might help to solve the differential equation for Y=y+1 as a function of X=x+1.

(ii) Find the solution $y(x)$ of the differential equation

$$\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x^2} + \exp(x) = 0$$

(iii) Find the solution $y(x)$ of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = 2\sin(3x)\,,$$

that satisfies $y(0) = 1,\ \dfrac{\mathrm{d}y}{\mathrm{d}x}(0) = 0\,.$

END OF PAPER

EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012 | Course EE1.1

Question Q1 | TOPIC General (long question) | Marks & seen/unseen

Parts

(i) $z^2 + z^{x^2} = -8$ find all roots

$2x^2 - 2y^2 = -8 \Rightarrow y = \pm\sqrt{x^2+4}$

4 seen similar

(ii) $(2-2i)^{2/3}$ in polar form

$(2-2i)^{2/3} = 8^{\frac{1}{3}} e^{-i\left(\frac{\pi}{4}+2\pi k\right)\frac{2}{3}}$

$= 2\, e^{-i\left(\frac{\pi}{6}+\frac{4}{3}\pi k\right)}$ $k = 0,1,2$

so: $2e^{-i\frac{\pi}{6}}$, $2e^{-i\frac{3\pi}{2}} = 2e^{i\frac{\pi}{2}}$

and $2e^{-i\frac{17}{6}\pi} = 2e^{-i\frac{5}{6}\pi} = 2e^{i\frac{7}{6}\pi}$

4 seen similar

$(2+2i)^{\frac{1}{3}}$ in polar form:

$(2+2i)^{\frac{1}{3}} = 8^{\frac{1}{6}} e^{i\left(\frac{\pi}{4}+2\pi k\right)\frac{1}{3}}$

$= \sqrt{2}\, e^{i\left(\frac{\pi}{12}+\frac{2k}{3}\pi\right)}$ $k = 0,1,2$

so: $\sqrt{2}e^{i\frac{\pi}{12}}$, $\sqrt{2}e^{i\frac{5}{4}\pi} = \sqrt{2}e^{-i\frac{5}{4}\pi}$

and $\sqrt{2}e^{i\frac{17}{12}\pi} = \sqrt{2}e^{-i\frac{7}{12}\pi}$

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question Q1	TOPIC	Marks & seen/unseen

Parts

(iii) Find q so that $\lim_{x\to\infty} e^{qx}\left(e^{\frac{2x}{3}} - (2+e^x)^{\frac{2}{3}}\right)$ is finite.

$$A = e^{\frac{2x}{3}}, \quad B = (2+e^x)^{\frac{2}{3}} \quad \text{and}$$

$$A - B = \frac{A^3 - B^3}{A^2 + AB + B^2} = \frac{e^{2x} - (2+e^x)^2}{A^2 + AB + B^2}$$

$$= -\frac{4e^x + 4}{A^2+AB+B^2} = -\frac{e^x}{e^{\frac{4x}{3}}} \cdot \frac{4 + 4e^{-x}}{1 + (2e^{-x}+1) + (2e^{-x}+1)^2}$$

$$\Rightarrow e^{qx}(A-B) = -e^{(q-\frac{1}{3})x} \cdot \frac{4 + 4e^{-x}}{1 + (2e^{-x}+1) + (2e^{-x}+1)}$$

For $q = \frac{1}{3}$ this converges to $-\frac{4}{3}$ as $x \to \infty$.

4 unseen

(iv) $$\lim_{x\to 0} \frac{\cos x - 1}{x^2} = \lim_{x\to 0} \frac{-\sin^2 x}{x^2(1+\cos x)} = -\frac{1}{2}$$

seen 4

(v) $$(\sin x)^{\sin x} = \exp(\sin x \ln \sin x)$$

$$\frac{d}{dx}(\sin x)^{\sin x} = (\cos x \ln \sin x + \cos x)(\sin x)^{\sin x}$$

$$= \cos x\,(1 + \ln(\sin x))\,(\sin x)^{\sin x}$$

unseen 4

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EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012 | Course EE1.1

Question Q1 | TOPIC | Marks & seen/unseen

Parts

(vi)

$$\int \frac{\sin x + 1}{\sin x - 1}\,dx = \int \frac{(\sin x+1)^2}{-\cos^2 x}\,dx$$

$$= \int -\frac{1}{\cos^2 x} - 2\frac{\sin x}{\cos^2 x} - \underbrace{\frac{\tan^2 x}{\frac{1}{\cos^2 x} - 1}}\,dx$$

$$= x - 2\tan x - 2\frac{1}{\cos x} + C' \quad (A)$$

4 seen similar

Alternatively

$$t = \tan\left(\frac{x}{2}\right), \quad \sin(x) = \frac{2t}{1+t^2}, \quad dx = \frac{2}{1+t^2}\,dt$$

$$\int \frac{\sin x + 1}{\sin x - 1}\,dx = x + 2\int \frac{1}{\sin x - 1}\,dx$$

$$= x + 2\int \frac{1}{\frac{2t}{1+t^2} - 1}\,\frac{2}{1+t^2}\,dt = x - 4\int \frac{dt}{(1-t)^2}$$

$$= x - 4\frac{1}{1-\tan\frac{x}{2}} + C' \quad (B)$$ (differs by a const from above)

$$(A) - (B) = 2$$

(vii)

$$\int_0^1 \frac{x}{(1-x^2)^{1/3}}\,dx = \frac{1}{2}\int_0^1 \frac{du}{(1-u)^{1/3}} = \frac{1}{2}\lim_{\varepsilon\to 0^-}\left[-\frac{3}{2}(1-u)^{2/3}\right]_0^{1+\varepsilon}$$

$$= \frac{1}{2}\left(-\frac{3}{2}(-1)\right) = \frac{3}{4}$$

4 unseen

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question Q1	TOPIC	Marks & seen/unseen

Parts

(viii) $f(x) = \frac{\ln(x-1)}{x}$ Taylor expansion: — 4 unseen

$$f'(x) = -\frac{\ln(x-1)}{x^2} + \frac{1}{x(x-1)}$$

$$f''(x) = 2\frac{\ln(x-1)}{x^3} - \frac{1}{x^2(x-1)} - \frac{2x-1}{x^2(x-1)^2}$$

$$f(2) = 0$$

$$f'(2) = \frac{1}{2}$$

$$\Rightarrow f(x) = \frac{1}{2}(x-2) + R_2(x)$$

with $R_2(x) = \frac{(x-2)^2}{2} f''(\xi)$, $\xi \in [2, x]$

(ix) $y' = \underbrace{\frac{y}{x} + 1}_{f(v)}$ homogeneous, $v = \frac{y}{x}$, $f(v) = v+1$ — 4 seen similar

$$\ln x = \int \frac{1}{f(v)-v}\, dv = v + c' \Rightarrow y = x(\ln x - c')$$

(x) $y'' + 4y' + 4y = \exp(x)$ — 4 seen similar

$a=1$, $b=4$, $c=4$ $\quad b^2 - 4ac = 0$ critical case

$$\lambda = -\frac{b}{2a} = -2$$

$$\Rightarrow y_{CF} = Ae^{-2x} + Bxe^{-2x}$$

$$y_{PI} = \frac{1}{9}\exp(x)$$

$$\Rightarrow y(x) = Ae^{-2x} + Bxe^{-2x} + \frac{1}{9}e^{x}$$

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question Q2	TOPIC Differentiation	Marks & seen/unseen

Parts

(i) $y = \dfrac{e^x}{e^{x^2}}$

$y' = \dfrac{e^x}{e^{x^2}} - 2x e^{x^2} \dfrac{e^x}{e^{2x^2}} = (1-2x)\, e^{x-x^2}$

3 seen similar

(ii) $y = \exp(\sin^{-1}(x))$

$y' = \dfrac{1}{\sqrt{1-x^2}} \exp(\sin^{-1}(x))$

3 unseen

(iii) $y = \ln(\cos(x^{\sinh x}))$

$= \ln(\cos(\exp(\sinh(x)\ln(x))))$

$y' = -\left(\dfrac{\sinh x}{x} + \cos(x)\ln(x)\right) \exp(\sinh(x)\ln(x))$

$\tan(\exp(\sinh(x)\ln(x)))$

$= -\left(\dfrac{\sinh x}{x} + \cos(x)\ln(x)\right) x^{\sinh x} \tan(x^{\sinh x})$

4 unseen

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question 2	TOPIC	Marks & seen/unseen

Parts

(iv) $\cos(y) = \sinh(x)$

$\Rightarrow \quad -y' \sinh(y) = \cos x$

$\Rightarrow \quad y' = \dfrac{-\cos x}{\pm\sqrt{1-\sinh^2(x)}} = \pm 1$

Award all marks for either sign

However if you use principle values it is -1, — 4 unseen

(v) $\dfrac{d^n}{dx^n}\left(x^2 \exp(\tfrac{1}{2}x)\right) = \sum_{i=0}^{n} \binom{n}{i} (x^2)^{(i)} \left(\exp(\tfrac{1}{2}x)\right)^{(n-i)}$ — 3 unseen

i	$(x^2)^{(i)}$	$\binom{n}{i}$
0	x^2	1
1	$2x$	n
2	2	$\frac{n(n-1)}{2}$
3	0	⋮
⋮	⋮	

$\left(\exp(\tfrac{1}{2}x)\right)^{(n-i)} = \left(\tfrac{1}{2}\right)^{(n-i)} \exp(\tfrac{1}{2}x)$

$\Rightarrow \dfrac{d^n}{dx^n}\left(x^2 \exp(\tfrac{1}{2}x)\right)$

$= e^{\frac{1}{2}x}\left\{\dfrac{x^2}{2^n} + \dfrac{nx}{2^{n-2}} + \dfrac{n(n-1)}{2^{n-2}}\right\}$ — 3

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EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012 | Course EE1.1

Question 3 | TOPIC Limits | Marks & seen/unseen

Parts

(i) $\lim_{x\to -1} \frac{(x-2)(x+2)}{(x-3)(x-1)} = \frac{-3}{8}$ — 3 seen similar

(ii) $\lim_{x\to 1} \frac{(x-2)(x^2-2)}{(x-3)(x+1)} = \lim_{x\to -1} 2\,\frac{(x-2)(x-1)}{(x-3)}$

$= 2\,\frac{6}{-4} = -3$ — 3 seen similar

(iii) $\sqrt{1+x} - \sqrt{x} = \frac{1}{\sqrt{1+x}+\sqrt{x}}$

$\Rightarrow \lim_{x\to\infty} \sqrt{x}\left(\sqrt{1+x}-\sqrt{x}\right) = \lim_{x\to\infty} \frac{\sqrt{x}}{\sqrt{1+x}+\sqrt{x}} = \frac{1}{2}$ — 4 seen similar

(iv) $\lim_{x\to 1} \frac{x^n-1}{x-1} = \lim_{x\to 1} \underbrace{x^{n-1}+x^{n-2}+\ldots+1}_{n \text{ terms}} = n$ — 5 seen similar

(v) $\lim_{x\to\pi} \frac{\sin(x)+(x-\pi)}{(x-\pi)^3}$ — 5 unseen

To use L'Hôpital, show that π is a root in numerator and denominator. Latter obvious for 0th, 1st, 2nd but not third derivative. Former obvious for 0th derivative, and then

1st $\quad \cos(x)+1 = 0 \quad$ at $x=\pi$

2nd $\quad -\sin(x) = 0 \quad$ at $x=\pi$

3rd $\quad -\cos(x) = 1 \quad$ at $x=\pi$

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question 3	TOPIC	Marks & seen/unseen
Parts	Third derivative of denominator: 6	

$$\Rightarrow \lim_{x \to \pi} \frac{\sin(x) + (x-\pi)}{(x-\pi)^3} = \frac{1}{6}$$

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question 4	TOPIC Integration	Marks & seen/unseen

Parts

(i) $\int_0^1 x \ln x \, dx$ integrand singular at $x=0$. — 2 unseen

$$\int x \ln x \, dx = \tfrac{1}{2}x^2 \ln x - \int \tfrac{1}{2} x \, dx = \tfrac{1}{2}x^2 \ln x - \tfrac{1}{4}x^2 + C'$$

$$\Rightarrow \int_0^1 x \ln x \, dx = \lim_{\varepsilon \to 0^+} \left[\tfrac{1}{2}x^2 \ln x - \tfrac{1}{4}x^2 \right]_\varepsilon^1$$

$$= -\tfrac{1}{4} \quad \text{. using } \lim_{x \to 0} x^2 \ln x = 0$$

Since $\lim_{x \to 0} x \ln x = 0$. — 3

(ii) $$\int_0^\infty x^n e^{-x} dx = \left[-x^n e^{-x} \right]_0^\infty + \int_0^\infty n x^{n-1} e^{-x} dx$$

$$= n \int_0^\infty x^{n-1} e^{-x} dx$$

5 unseen

(iii) $$\int \cos^3(x) \, dx = \int (1 - \sin^2(x)) \cos(x) \, dx$$

$$= \sin(x) - \tfrac{1}{3}\sin^3(x) + C'$$

4 seen similar

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EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012 | Course EE1.1

Question 4 | TOPIC | Marks & seen/unseen

Parts

(iv) $$\int \frac{x^2-1}{x^3+2x^2-3x}\,dx = I$$

Denominator: $x^3+2x^2-3x = x(x^2+2x-3) = x(x+3)(x-1)$

Numerator: $(x+1)(x-1)$

$$\Rightarrow I = \int \frac{x+1}{x(x+3)}\,dx$$

$$\frac{x+1}{x(x+3)} = \frac{A}{x} + \frac{B}{x+3} \quad \Rightarrow A = \tfrac{1}{3} \quad B = \tfrac{2}{3}$$

3 seen similar

$$\Rightarrow I = \tfrac{1}{3}\int \frac{1}{x} + \frac{2}{x+3}\,dx = \tfrac{1}{3}\ln|x| + \tfrac{2}{3}\ln|x+3| + C'$$

3

(v) $$\int \frac{dx}{\sqrt{8+2x-x^2}} = I$$

$$8+2x-x^2 = -(x-4)(x+2)$$

seen similar

$$y = x-1$$

$$\Rightarrow 8+2x-x^2 = -(y-3)(y+3) = 9-y^2$$

$$I = \int \frac{1}{3}\frac{dy}{\sqrt{1-\frac{y^2}{9}}} = \sin^{-1}\left(\frac{y}{3}\right) + C'$$

$$= \sin^{-1}\left(\frac{x-1}{3}\right) + C'$$

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question 5	TOPIC Complex numbers + Taylor series	Marks & seen/unseen

Parts

(i) $|z+1| = 2^* + z \quad (z = x+iy) \Rightarrow (x+1)^2 + y^2 = 4x^2$

$\Rightarrow y^2 = 3x^2 - 2x - 1$

$y = \pm\sqrt{3x^2-2x-1}$ for $3x^2-2x-1 \geq 0$, $x \in \mathbb{R}$

Roots of $3x^2-2x-1$ by inspection: $x=1$

$\Rightarrow 3x^2-2x-1 = 3(x-1)(x+\frac{1}{3})$

other root: $x = -\frac{1}{3}$

$\Rightarrow x \geq 1$ and $x \leq -\frac{1}{3}$, $x \in (-\infty, -\frac{1}{3}] \cup [1, \infty)$ — 5 seen similar

$\arg(z(z-2)) = \pi \Rightarrow$

$z(z-2) = -r$ with $0 \leq r \in \mathbb{R}$

Where is $z(z-2)$ negative?

$z \in (0,2) \subset \mathbb{R}$, $z = 1 \pm i\sqrt{\frac{r}{2}-1}$ for $r > 2$ — seen similar

(ii) $1+2i = \sqrt{5}\, e^{i \tan^{-1}(2)}$ — 2 seen similar

$(\sqrt{3}+i)^{\frac{1}{3}} = 2^{\frac{1}{3}} e^{i \underbrace{\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)}_{\frac{\pi}{6}} \frac{1}{3} + i 2\pi \frac{k}{3}}$ $\quad k = 0, 1, 2$ — 3 seen similar

$2^{\frac{1}{3}} e^{i\frac{\pi}{18}}$, $2^{\frac{1}{3}} e^{i\frac{13\pi}{18}}$, $2^{\frac{1}{3}} e^{i\frac{25\pi}{18}} = 2^{\frac{1}{3}} e^{-i\frac{11\pi}{18}}$

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	EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012	Course EE1.1
Question 5	TOPIC	Marks & seen/unseen

Parts

(iii) $\sin(4\theta) = \mathrm{Im}\left(e^{i4\theta}\right) = \mathrm{Im}\left((\cos\theta + i\sin\theta)^4\right)$

$= 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$ — 4 seen similar

(ii) $\int \cos(x)\, e^{2x}\, dx = \int \frac{1}{2}\left(e^{ix} + e^{-ix}\right)e^{2x}\, dx$

$= \frac{1}{2}\frac{1}{2+i}e^{(2+i)x} + \frac{1}{2}\frac{1}{2-i}e^{(2-i)x} + C'$ — 3 unseen

$= \frac{1}{2}e^{2x}\frac{1}{5}\left\{(2-i)e^{ix} + (2+i)e^{-ix}\right\} + C'$

$= \frac{1}{5}e^{2x}\left\{2\cos(x) + \sin(x)\right\} + C'$ — 3

Alternatively integration by parts twice:

$\int \cos(x)\, e^{2x}\, dx = \sin x\, e^{2x} - 2\int \sin(x)\, e^{2x}\, dx$

$= \sin x\, e^{2x} + 2\cos x\, e^{2x} - 4\int \cos(x)\, e^{2x}\, dx$

$\Rightarrow \int \cos(x)\, e^{2x}\, dx = \frac{1}{5}e^{2x}\left\{2\cos(x) + \sin(x)\right\} + C'$

(v) $f(x) = x\ln(x)$

$f'(x) = \ln(x) + 1$

$f''(x) = \frac{1}{x}$

$f'''(x) = -\frac{1}{x^2}$

$f(x) = (x-1) + \frac{1}{2}(x-1)^2 + R_3(x)$

$R_3(x) = \frac{(x-1)^3}{6}\left(-\frac{1}{\xi^2}\right)$

$\xi \in [1, x]$

unseen

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EXAMINATION ~~QUESTIONS~~/SOLUTIONS 2011-2012 | Course EE1.1

Question 6 | TOPIC ODEs | Marks & seen/unseen

Parts

(i)

$$y' = \frac{y^2+x^2+2(x+y)+2}{2x^2+4x+2} = \frac{(y+1)^2+(x+1)^2}{2(x+1)^2}$$

$X = x+1 \quad Y = y+1 \quad Y' = \frac{Y^2+X^2}{2X^2} = \underbrace{\frac{1}{2}(v^2+1)}_{f(v)}$

with $v = \frac{Y}{X}$

3 seen similar

$$\Rightarrow \ln X = \int \frac{dv}{f(v)-v} = \int \frac{2}{(1-v)^2}\,dv = \frac{2}{1-v} + C'$$

$$\frac{1}{2}(\ln X - C') = \frac{1}{1-v}$$

$$v = 1 - \frac{2}{\ln X - C'}$$

$$\Rightarrow y = (x+1)\left\{1 - \frac{2}{\ln(x+1) - C'}\right\} - 1$$

4

(ii)

$$\frac{1}{x}y' + \frac{y}{x^2} + \exp(x) = 0 \Rightarrow$$

$$y' + \frac{y}{x} = -xe^x \quad \text{linear} \quad R = e^{\int \frac{1}{x}dx} = x$$

$$\Rightarrow y(x) = \frac{-1}{x}\int x^2 e^x dx$$

6 seen similar

$$\int x^2 e^x dx = x^2e^x - 2\int xe^x dx = x^2e^x - 2xe^x + 2e^x + C$$

$$\Rightarrow y(x) = -e^x\frac{x^2-2x+2}{x} + \frac{C'}{x}$$

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Question 6 | TOPIC | Marks & seen/unseen

Parts (iii)

$$y'' + 2y' - 3y = 2\sin(3x)$$

$a=1,\ b=2,\ c=-3 \qquad b^2 - 4ac = 16 > 0$

$$\lambda_1 = \frac{-2+\sqrt{16}}{2} = 1 \qquad \lambda_2 = \frac{-2-\sqrt{16}}{2} = -3$$

$$y_{CF} = Ae^{x} + Be^{-3x}$$

2 seen similar

$$y_{PI} = \alpha\sin(3x) + \beta\cos(3x)$$

$$y_{PI}' = 3\alpha\cos(3x) - 3\beta\sin(3x)$$

$$y_{PI}'' = -9\alpha\sin(3x) - 9\beta\cos(3x)$$

$$y_{PI}'' + 2y_{PI}' - 3y_{PI} = -9\alpha\sin(3x) - 9\beta\cos(3x) - 6\beta\sin(3x) + 6\alpha\cos(3x) - 3\alpha\sin(3x) - 3\beta\cos(3x)$$

$$-9\alpha - 6\beta - 3\alpha = 2 \Rightarrow -3\beta - 1 = 6\alpha$$

$$-9\beta + 6\alpha - 3\beta = 0 \Rightarrow -9\beta - 3\beta - 1 - 3\beta = 0$$

$$\beta = -\frac{1}{15}$$

$$\alpha = -\frac{2}{15}$$

$$\Rightarrow y_{PI}(x) = -\frac{2}{15}\sin(3x) - \frac{1}{15}\cos(3x)$$

2

$$y(x) = Ae^{x} + Be^{-3x} - \frac{2}{15}\sin(3x) - \frac{1}{15}\cos(3x)$$

$$y(0) = A + B - \frac{1}{15} = 1 \Rightarrow A + B = \frac{16}{15}$$

$$y'(x) = Ae^{x} - 3Be^{-3x} - \frac{2}{5}\cos(3x) + \frac{1}{5}\sin(3x)$$

$$y'(0) = A - 3B - \frac{2}{5} = 0 \Rightarrow \frac{16}{15} - 4B - \frac{2}{5} = 0 \Rightarrow B = \frac{1}{6}$$

$$A = \frac{16}{15} - \frac{1}{6} = \frac{32}{30} = \frac{9}{10}$$

3